

Terrain and 3D buildings

How to get the terrain and buildings of any site in the world, prepare them in QGIS, then use them in Illustrator, AutoCAD, Rhino, Blender and Archicad. A detailed track for France and the EU, where data is very rich, and another for Vietnam, where you have to combine several sources.

EPSG:2154 Lambert 93 EPSG:32648 UTM 48N · Hà Nội EPSG:4326 WGS 84

Updated: September 2026

00

The theory: GNSS, geodesy, GIS

GPS, GNSS: positioning with satellites

GNSS

GPS

GPS
GLONASS
Galileo

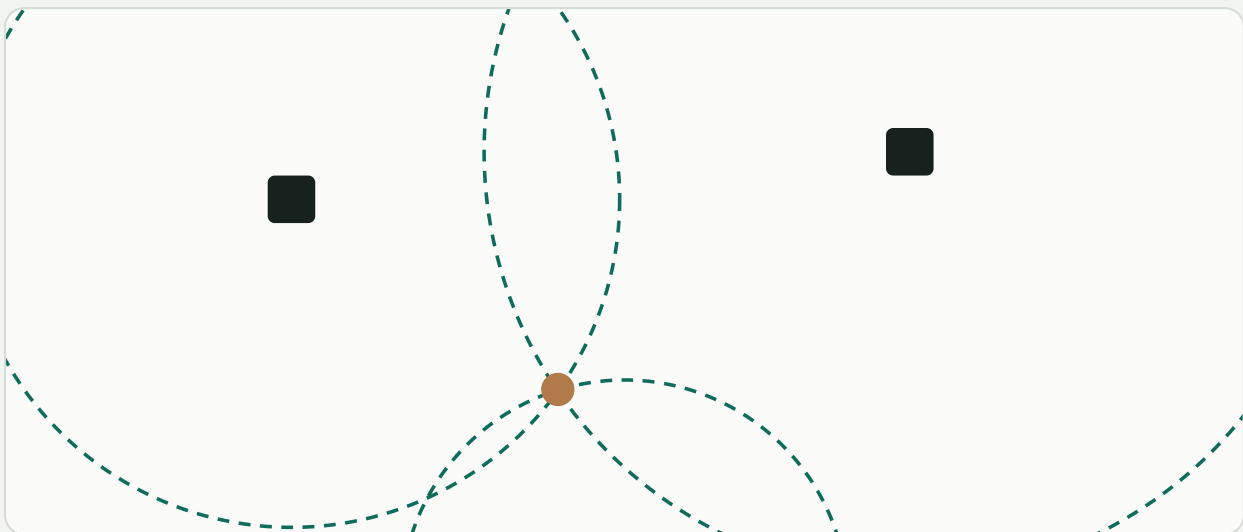
BeiDou

QZSS ·
NavIC

How it works: trilateration

distance

at least 4 satellites



one nanosecond

30 cm

Accuracy in practice

Smartphone

SBAS (EGNOS,
WAAS...)

PPP / Galileo
HAS

RTK

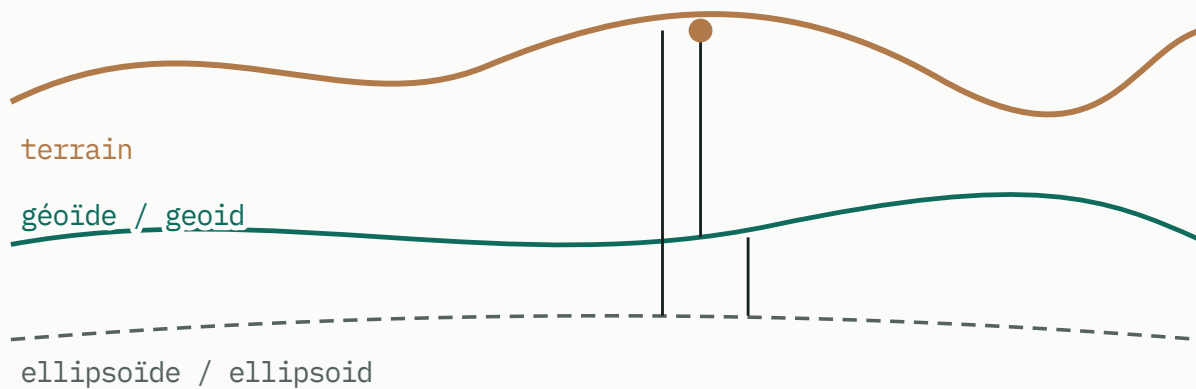
Geodesy: the shape of the Earth

ellipsoid

geodetic datum

- WGS 84
- ETRS89 / RGF93
- VN-2000

geoid



$$h = H + N$$

vertical datum

Coordinates and projections

Geographic coordinates

21.0287° N, 105.8524° E

-
-
-

Projected coordinates

projection

- angles
- areas
-

EPSG:3035

- UTM
- Lambert 93
- Web Mercator EPSG:3857

- EPSG

GIS: software that links maps and data

GIS

store, analyse and represent

attribute table

Vector

Raster

A little history (useful for a thesis)

- 1854, John Snow
- 1960s, Roger Tomlinson
- 1969, Ian McHarg
- 2004, Steve Coast

Remote sensing: measuring from afar

Optical (passive)

Radar (active)

LiDAR

inertial measurement unit

Web mapping standards

- OGC
- WMS / WMTS
- WFS
- Tiles

Further reading

- _____
- _____
- _____
- _____
- _____
- _____
- _____
- _____

01

Plane coordinates, step by step: UTM

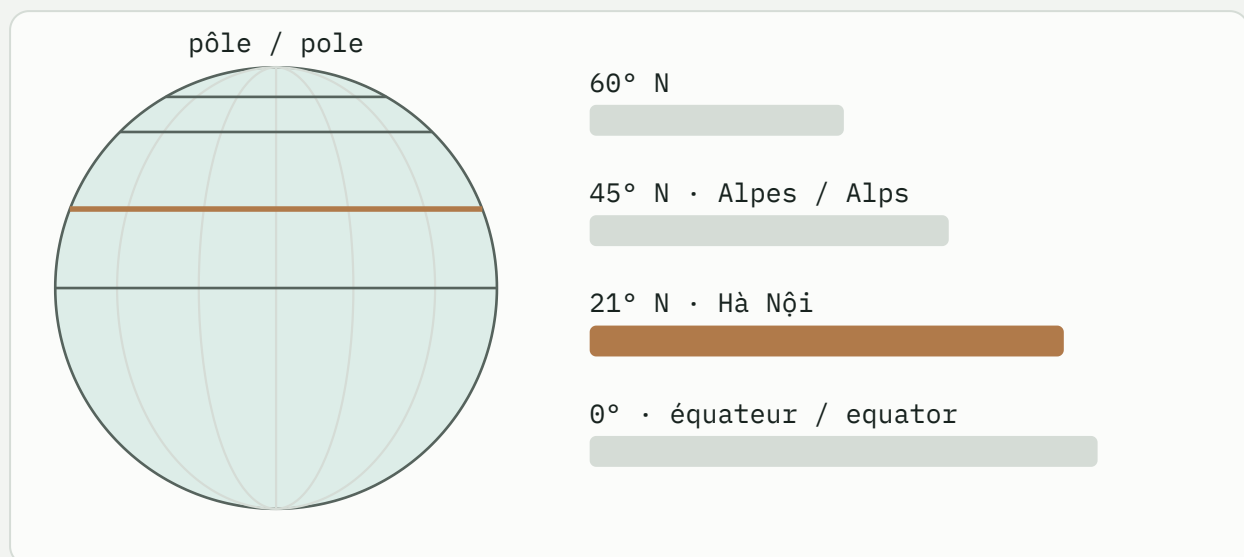
EPSG:32648

588,574;

2,325,560

1. The problem: a degree has no fixed length

degrees



one degree of longitude

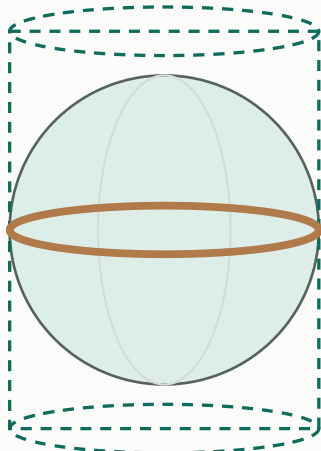
project
rectangular

plane

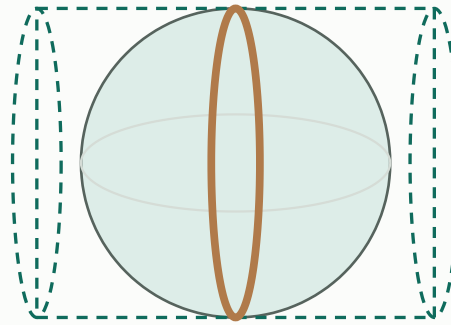
2. The idea: wrap the Earth in a cylinder

touches

lay the cylinder on its side



Mercator



Mercator transverse (UTM)

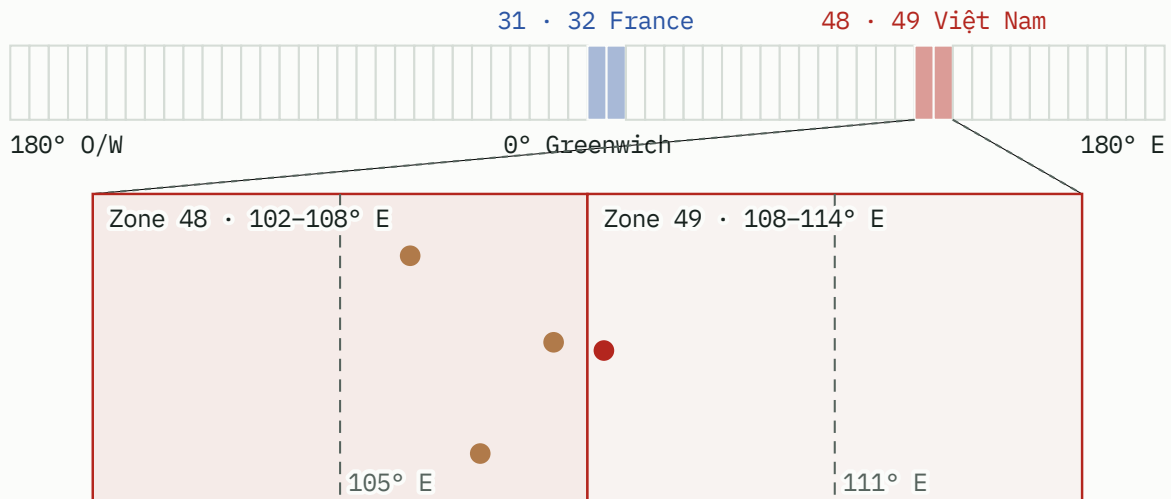
equator

meridian

3. Sixty zones of 6°

60 vertical zones

central meridian



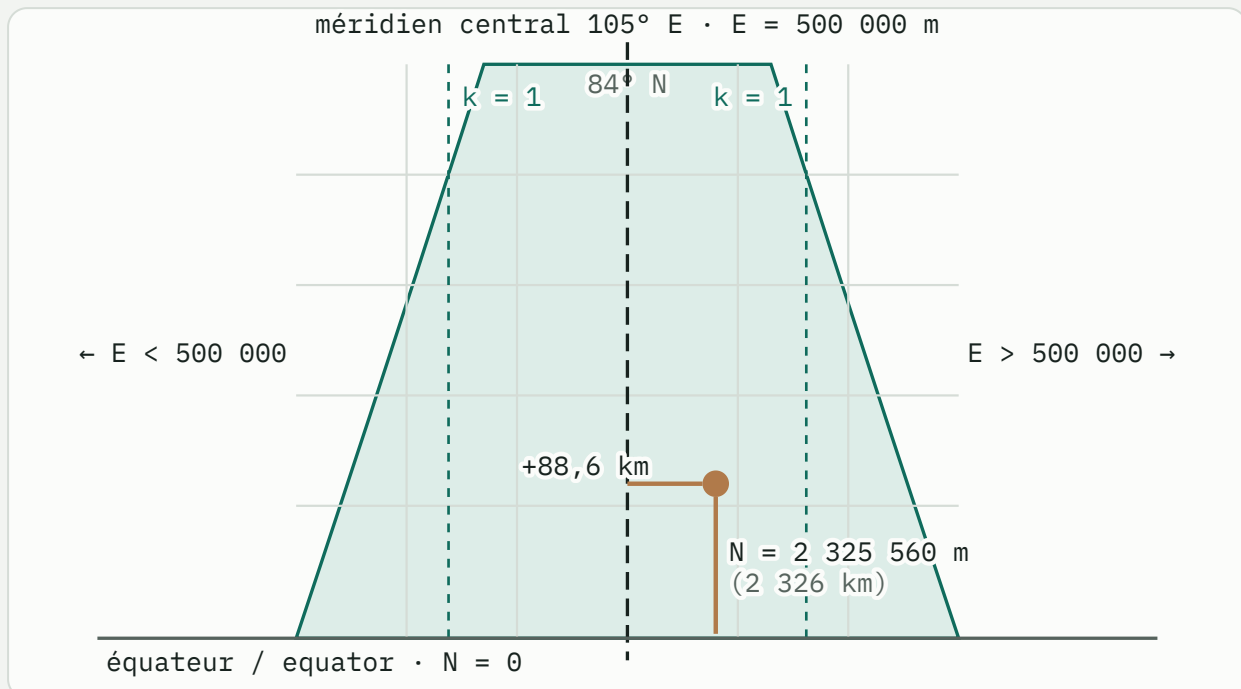
$\text{zone} = \text{floor}((\text{longitude} + 180) \div 6) + 1$

zone 48

$\text{zone} \times 6 - 183$

4. A zone's grid: Easting and Northing

- E, Easting
500,000 m
- N, Northing



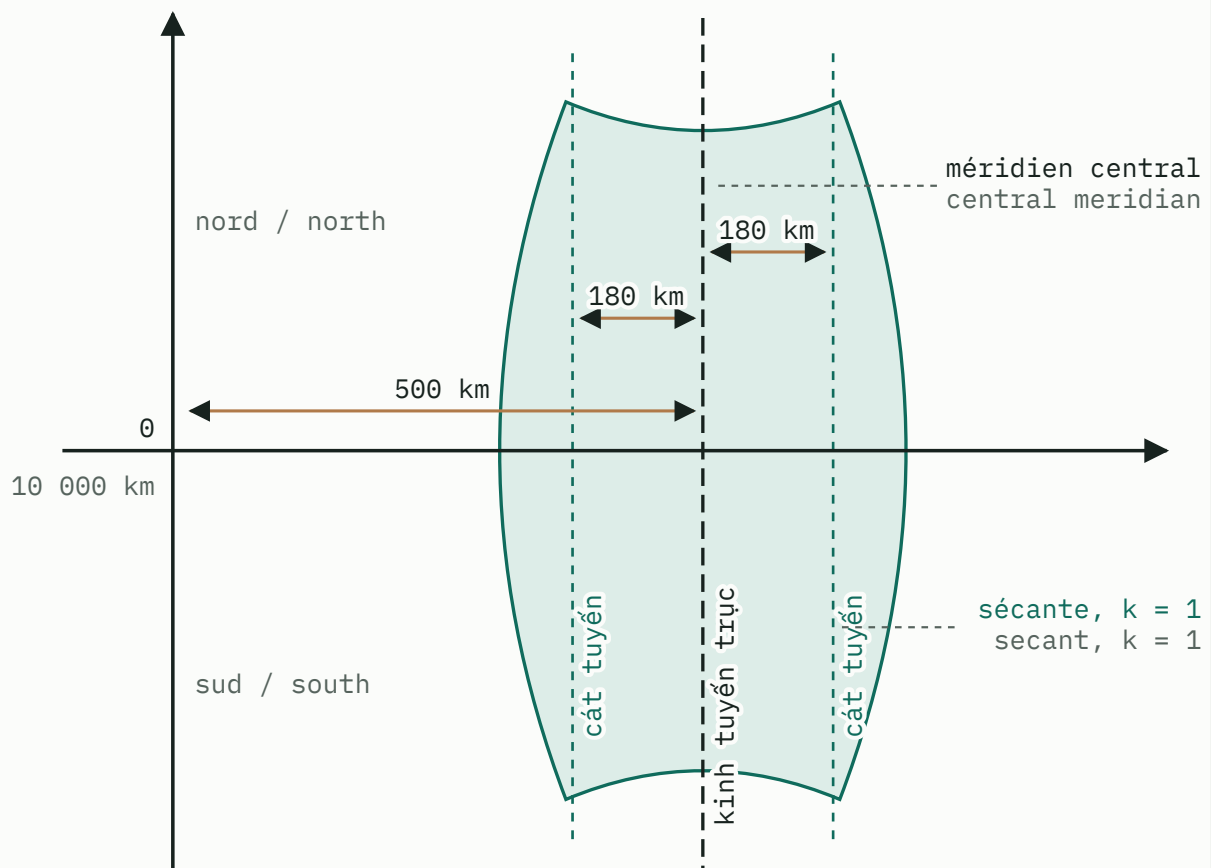
rectangular

588,574 m

The diagram from Vietnamese surveying courses (x = North!)

up
means East

one notation difference that trips everyone
x means North and y



x (N)

y (E)

central meridian

secant lines

• 180 km

500 km

false easting

0 • 10 000
km

false northing

Curved
edges

then East

North first,

Explore the zone with your mouse

Interactive exercise: reading the diagram

5. Worked example: Hoan Kiem Lake

21.0287° N; 105.8524° E



The zone

48



Offset from the central meridian

88.6 km



The Easting

588,600 m



The Northing

2,325,800 m

Reading a UTM coordinate



48

N northern

EPSG:32648

EPSG:32748

E

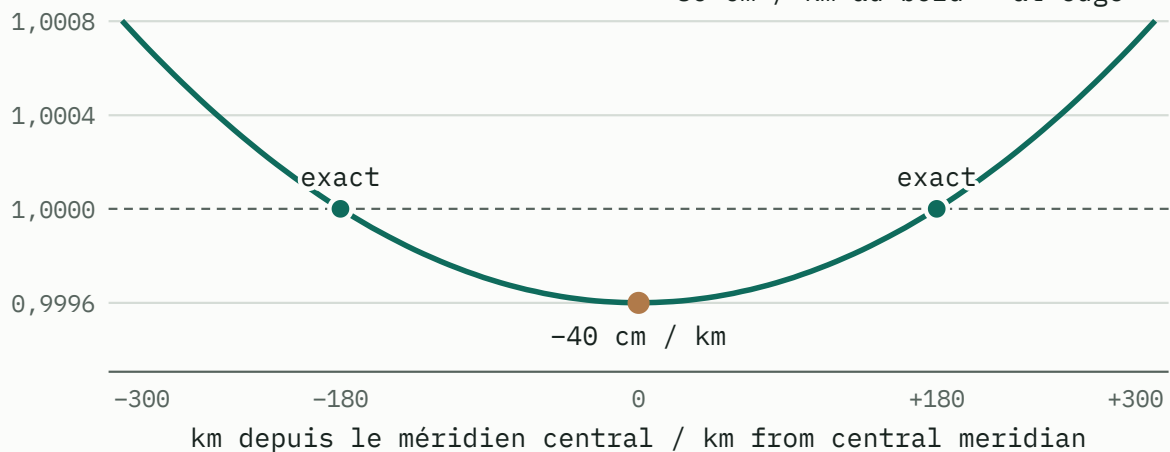
N

6. The small price: scale and north

0.9996

k (facteur d'échelle / scale factor)

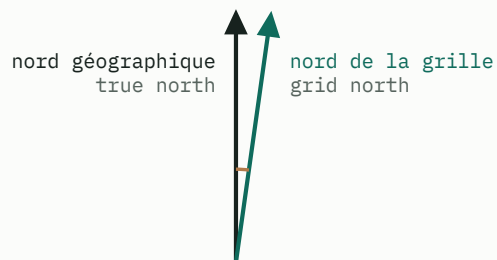
+80 cm / km au bord · at edge



negligible for an

architecture project

Grid north $\neq$ true north



convergence

0.3°

grid

Mind the zone boundaries

one

never

7. Why it's perfect for an architect

- 1 unit = 1 metre.

- Perpendicular axes.

$$+ (N_2 - N_1)^2$$

$$d = \sqrt{(E_2 - E_1)^2}$$

- Correct areas.

- Recentring becomes simple

$$X_0 = 588,000; Y_0 = 2,325,000$$

8. Other plane systems: same logic

UTM 48N

VN-2000

Lambert 93

Web
Mercator

9. Check your understanding

▼ Which UTM zone is Da Nang in (108.20° E)?

zone 49

▼ A point has E = 471,000 m. Is it east or west of the central meridian, and how far?

west

29 km

▼ Two corners of a plot: (588,520; 2,325,480) and (588,560; 2,325,510). What's the frontage length?

50 m

▼ Why do your OSM buildings and your DEM show up 800 km apart in Rhino?

before

Further reading

- _____
- _____

•

The basics

DTM	terrain
DSM	surface
nDSM / CHM	height
LiDAR point cloud	.las .laz .copc.laz
Building footprints	
LoD1 / LoD2	
Photoreal mesh	

Resolution: the number that matters

≤ 1 m

Plot / block

1 - 10 m

District / city

30 m

Territory / landscape

Coordinate reference systems (CRS)

EPSG

EPSG:4326

projection

- **FR** EPSG:2154

- **UE** EPSG:25832

- **VN** EPSG:32648

EPSG:32649

VN-2000

- **MONDE**

03

The 7-step method

○ Define the area

margin

.geojson

○ Pick the project CRS

○ Download terrain

○ Download buildings



Assemble in QGIS



Move near the origin



Export to your software

04

Which source for which case

FRANCE

UE

VIETNAM

MONDE

Global sources

Terrain (DTM / DSM)

Copernicus DEM

GLO-30

FABDEM

DTM

ALOS AW3D30

NASADEM / SRTM

GEBCO

OpenTopography

Buildings and urban context

OpenStreetMap

QuickOSM

height

building:levels

GlobalBuildingAtlas

heights LoD1

Google Open
Buildings

Temporal

2.5D

Microsoft Building
Footprints

Overture Maps

overturemaps

Google Photorealistic
3D Tiles

Base imagery (orthophotos, satellite)

- _____
- QuickMapServices

France and the European Union

FR IGN datasets

LiDAR HD —
point clouds

.copc.laz

MNT / MNS /
MNH LiDAR
HD

50 cm

RGE ALTI

BD TOPO

height

BD ORTHO

Cadastre

Pan-European

- _____
- _____
- _____

Very well covered (LiDAR + LoD2)

- Pays-Bas / NL _____
- Suisse / CH _____
- Allemagne / DE _____
- Espagne / ES _____
- Danemark, Finlande, Pologne, Flandre _____

Patchy coverage

- Italie / IT _____

- _____

Vietnam

no public LiDAR

cross-check

verify on site

Terrain

- FABDEM
- Copernicus GLO-30 / ALOS
- _____

Buildings

- Google Open Buildings 2.5D
- GlobalBuildingAtlas
- OpenStreetMap
- Microsoft / Overture

Documents and context

- _____
- _____
- _____

Make your own survey

- Ground photogrammetry

- iPhone/iPad Pro with LiDAR
- Heights

08

One-click tools

CADMAPPER

swzpln.de

Blosm (Blender)

BlenderGIS

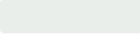
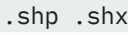
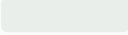
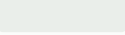
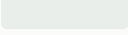
TouchTerrain

Heron / Elk
(Grasshopper)

Handling the files in QGIS

LTR

Formats you will meet

 GeoTIFF	
 	
  	
 GeoPackage	
 Shapefile	 .shp .shx  .dbf .prj
  	
 CityGML ·  CityJSON	
    	

QGIS plugins to install

- QuickOSM
- OpenTopography DEM Downloader
- QuickMapServices
- Qgis2threejs

Key operations

○ Set the project CRS

2154

32648

○ Merge tiles

○ Reproject and clip

○ Contours and hillshade

○ Make contours 3D

○ Fill building heights

h

```
coalesce("height", "building:levels" * 3.3, 9)
```

○ Point clouds

.copc.laz

.ply

.e57

.obj

○ Export

Organise a project folder

```
site_hanoi_hoankiem/  
├─ 00_zone/          emprise.geojson · README.txt (sources, dates, licences, CRS  
├─ 01_brut/          fichiers téléchargés, jamais modifiés / raw downloads, neve:  
├─ 02_qgis/          projet.qgz · donnees.gpkg (reprojeté, découpé / reprojected  
├─ 03_export/        topo_EPSG32648.dxf · plan.svg · contexte.obj · terrain.tif  
└─ 04_logiciels/     .pln · .3dm · .blend · .ai
```

origin point

```
X0 = 585000; Y0 = 2324000
```

Into architecture software

Illustrator

For:

fixed scale

AutoCAD

For:

MOVE

Civil 3D

Map 3D

MAPIMPORT

Rhino (+ Grasshopper)

For:

metres

Patch

MeshPatch

Heron

h

Heightfield

.e57

.ply

.xyz

Contour

Blender

For:

BlenderGIS

h

Blosm

Archicad

For:

Mesh

.obj

.xyz

.3dm

.e57

**Grasshopper–
Archicad Live Connection**

Which format for which tool

Illustrator

AutoCAD

Rhino

Blender

Archicad

Three full recipes

▼ VN Recipe A — A block in Hanoi, for Archicad and Blender

EPSG:32648

building highway natural=water

h

h

▼ FR Recipe B — A district in France, for Rhino

LiDAR HD DTM

BD TOPO

EPSG:2154

batiment

hauteur

hauteur

.e57

▼ MONDE Recipe C — A large landscape, for Illustrator and a physical model

Contour

Common pitfalls and licences

Mistakes that cost you a day

- DSM mistaken for DTM
- Mixed CRS
- Huge coordinates
- Vertical datum
- 8-bit heightmap
- Too much data
- Satellite heights
- Data date

Credit sources on your boards

OpenStreetMap

IGN

Copernicus DEM

FABDEM

Google Open
Buildings

GlobalBuildingAtlas,
Overture, EUBUCCO

Google 3D Tiles

Glossary FR / EN / VI

Relief, topographie

Modèle numérique de terrain (MNT)

Modèle numérique d'élévation

Courbes de niveau

Nuage de points

Emprise de bâtiment

Hauteur de bâtiment

Système de coordonnées

Cadastre

Orthophoto

Plan masse

Plan d'urbanisme

Maison-tube

Relevé

Photogrammétrie

Guide written for architecture students at HAU (Hanoi) · September 2026. Online services change: if a link breaks, search for the source's name.